

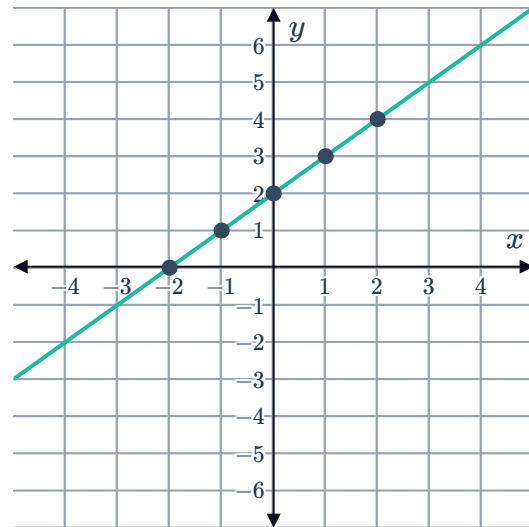
Evaluating functions



- 1 Consider the function $y = 4x - 3$.
 - a Identify the independent variable of the function.
 - b Identify the dependent variable of the function.
- 2 For a function f , state what $f(3)$ represents.
- 3 Suppose that $(7, -6)$ is an ordered pair that satisfies the function g . Write this situation using function notation.

- 4 Consider the following graph of the function $f(x)$.

- a Find $f(0)$.
- b Find $f(-2)$.
- c Find the value of x such that $f(x) = 3$



- 5 Consider the function $f(x) = 8x + 6$.
Determine the output produced by the input value $x = 5$.
- 6 Consider the function $f(x) = -9x^2 - 8x - 4$.
Determine the output produced by the input value $x = -5$.
- 7 If $f(x) = 4x + 4$, find:
 - a $f(2)$.
 - b $f(-5)$.
- 8 If $f(x) = 3x - 1$, find:
 - a $f(3)$
 - b $f(-4)$



9 If $g(x) = \frac{7x}{4}$, find:

a $g(5)$

b $g(-4)$

10 If $f(x) = \frac{2x-3}{5}$, find:

a $f(4)$

b $f(-5)$

11 Consider the function $f(x) = 2x^2 - 2x + 5$.
Evaluate $f\left(\frac{1}{2}\right)$.

12 Consider the function $f(x) = x^2 + 8x$.

a Form an expression for $f(a)$.

b Form an expression for $f(b)$.

13 Consider the function $p(x) = x^2 + 8$.

a Evaluate $p(2)$.

b Form an expression for $p(m)$.

14 Consider each of the following equations.

i Rewrite the equation using function notation $f(x)$.

ii Find the value of $f(3)$.

a $x + 3y = 6$

b $-6x + 5y = 7$

c $y + 6x^2 = 3 - x$

15 Consider the equation $x - 4y = 8$.
Assume that y is a function of x .

a Rewrite the equation using function notation $f(x)$.

b Find the value of $f(12)$.

16 Consider each of the following equations.



- i Rewrite the equation using function notation $f(x)$.
- ii Find the value of $f(2)$.

a $5x - 3y = 7$

b $y - 4x^2 = 5 + x$

17 Find the value of x that satisfies the equation $\frac{1}{4} = \frac{x + 12}{8}$.

18 Consider the function $f(x) = -4x^2 - 5x + 6$.
Determine $f(-5)$.

19 State whether each of the following is true or false.

- a The cost of a pizza depends on the size of the pizza, so the cost of a pizza is a function of the size of the pizza.
- b The size of a pizza depends on the price of the pizza, so the size of a pizza is a function of the price of a pizza.

20 The model $P(t) = 3500 \times 2^t$ describes the population of fish in a lake after t years, where $t = 0$ is the initial time when the population of fish was first recorded.
Determine and interpret the correct value of $P(2)$.

Additional questions

21 Sharon works at a pizza shop and makes \$15 per hour. Her wages can be represented by the function $W(t) = 15t$, where t represents the number of hours that she works.

- a Explain what the expression $W(19)$ represents.
- b Explain what the equation $W(n) = 165$ represents.
- c Find her wages after working 26 hours.

22 Mindy can model her tomato plant's height in inches over time using the function $h(t) = 2^t$, where t is the number of weeks after Mindy planted it.

- a Explain what the equation $h(4) = 16$ represents.
- b Find the height of Mindy's tomato plant after 6 weeks.

23 Consider the function $f(x) = ax + b$. If $f(4)$  and $f(-1) = -3$, find the values of a and b .
Explain how you arrive at the answer.

24 Describe a scenario that can be represented by the function $f(x) = \frac{120}{x} + 5$.